First Name: Fred
Last Name: Ayres
CID: 02657015
Group Number: 14

ELEC40006-Electronics Engineering Design Project 1

Reflection on Professional Practice Form

Please answer all of the following questions in a total of 300 words (+10%). This is a compulsory component of the module and must be passed.
Include your name, surname, and CID number on the form.
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All completed forms must be compiled and attached to the end of your group's project report.
Submit the full report, including all reflection forms, by 17:00pm on 11 June 2026.

1. How effectively did your group function as a team? How effective was your personal contribution to the team?
2. How did a diversity of people in the teams help you to perform better as a team?
3. What skills and knowledge did you acquire in this project? How?

1. I am pleased that everyone contributed to at least one sensing circuit, although some people were able to contribute more to the team. I think we made a good effort to distribute the work fairly given our time constraints. For example, when one of us was struggling with one of the sensor circuits, other people in the team volunteered to help such that we would deliver on time. Personally, I worked on research and testing of the Magnet sensor, Design and Implementation of the IR circuit, Power System Design and Strip Board Construction.

2. A diversity of people in the teams allowed everyone to work on different sections on the project, working at their strengths and limitations.

3. I significantly improved my soldering skills as I was responsible for soldering the entire stripboard. I improved my proficiency in designing detection circuits where I had to consider different sources of noise (e.g. ambient DC noise from room lighting or Johnson Noise due to a very large gain stage in our circuit), find ways to reduce it: filter circuits. I also had to consider the gain in my circuit given the limitations: I wanted to be able to detect my rock at least 3cm away but if my gain was too large it would produce a lot of noise. Therefore, I also had to assist in researching a new IR sensor with a larger angle of detection because our previous sensor had a limited field of view. As we initially planned to use different drivers (the WPSH207 driver shield) for our circuit, I was responsible for researching and designing the new power system. Although, in the end due to our shield not arriving in time, we ended up using the drivers given to us.

First Name: Sijie
Last Name: Fei
CID: 02686898
Group Number: 14

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3. What skills and knowledge did you acquire in this project? How?

1. Our group of six divided into hardware and software streams. I owned the entire firmware and web dashboard: UART age decoding, IR classification using a Poisson log-likelihood ratio test, ultrasonic and magnetometer processing, multi-sensor fusion, mecanum motor control, and the real-time operator interface. Teammates designed the analogue front-ends — radio receiver, IR detector, ultrasonic interface, and motor driver wiring. We agreed a digital interface contract early (signal types, pin assignments, voltage levels), enabling both streams to develop in parallel. I built and tested the full classification pipeline in simulated mode before any hardware existed, so integration became a matter of flipping a compile flag. When the radio receiver turned out to have inverted logic levels, having working software made the fault easy to isolate. My contribution was substantial in scope but narrow in discipline; attending more hardware bench sessions would have helped me catch interface issues earlier.

2. Our team spanned different nationalities and academic strengths. Members with stronger analogue-electronics backgrounds designed robust sensor circuits, while my programming experience enabled the classification algorithms and dashboard. One teammate with mechanical experience led the chassis modifications for mecanum wheels. This range meant no single point of failure — someone could always offer a different perspective when another was stuck. Communicating across disciplines also forced precise interface documentation, which improved overall system quality.

3. I learned ARM embedded programming on the SAMD21, non-blocking real-time architecture, and HTTP server design under severe resource constraints — the WiFi101 module supports only four concurrent sockets. Implementing the Poisson LRT taught me to apply statistical decision theory to a real classification problem. The dashboard required asynchronous JavaScript, touch-event handling, and responsive design. Most valuably, debugging a motor-control watchdog failure taught me that embedded networking differs fundamentally from desktop networking: latency and concurrency assumptions that hold on a laptop collapse on a microcontroller. These lessons came through iterative development and integration failures no textbook could replicate.

First Name: Zeyu
Last Name: Liu
CID: 02590274
Group Number: 14

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3. What skills and knowledge did you acquire in this project? How?

1. Overall, our group worked reasonably. We divided tasks based on each member's strengths and interests, communicated about the main design issues, and combined our work into the final project and report. However, the teamwork was not perfect, as some members contributed less consistently than others, creating uneven workload and slowing down progress. My personal contribution was mainly in PCB design/construction/testing, ultrasonic signal detecting circuit, circuit prototyping/debugging, and presentation/report writing. I was able to complete my work efficiently and effectively.

2. The diversity within the team helped because different members had different strengths and approaches. Some focused more on circuit behavior and component choices, while others considered layout, testing, assembly, or documentation. This helped the active members identify problems from more than one perspective. For example, different opinions on sensor placement, component choice, and PCB manufacturability were considered along the PCB design. However, the benefit of diversity was limited by the unequal level of participation across the group.

3. I gained practical knowledge of PCB design, sensor integration, circuit design, circuit construction, manufacturing preparation, and circuit/ PCB debugging. I developed a better understanding of how an engineering design project moves from an initial idea to a practical working system. I improved my ability to apply technical knowledge to real designs, rather than only solving isolated theoretical problems. I also improved skills in documentation, and group communication. All above were developed through working on the design, solving practical problems during the project, responding to issues found during testing, and contributing to the project consistently and showing up everyday.

First Name: Owais
Last Name: Darwesh
CID: 02568029
Group Number: 14

ELEC40006-Electronics Engineering Design Project 1

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1. I think overall our group functioned well as a team. Everyone was very early split off into the sub-modules we decided early on. It did mean working in isolation for periods of time, however I think it suited many members of our team, hence this approach worked. When it came to integration time, where more teamwork would be required, our team mobilised reasonably well. We were able to get the work done which was the main thing, however we lacked the structure we had from the get-go. The work-balance became more uneven across members, which is something we should have probably worked on. As for my own contribution, one of my more collaborative roles was as the report editor. I had the task of getting everyone reading the particular sections they worked on, suggest edits, and from there improve the report. I think this iterative approach really helped the team as they were clued in more on what was going on.

2. The diversity of people really played in to our hands. We were able to utilise specific members' strengths in their particular field, whether it be programming or ADC. This allowed the team to make progress in lots of different areas simultaneously, which meant we had more time than others for integration. For example, whilst the electrical engineers designed their circuits, they didn't have to worry about the software, which I believe saved time.

3. I have found working on the project that, whilst it has built up some academic skills around problem solving, designing your own tasks, allowing a different perspective to a problem etc., I found that I mainly improved my soft skills. At times, I had to recognise that while my solution was correct, this may not be easy for someone else to work with. This was one of the reasons we digitised all our signals, so that software team could have an easier time managing the telemetry.

First Name: Ximan
Last Name: Wang
CID: 02374979
Group Number: 14

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1. Overall, our group functioned well together. We divided responsibilities early on according to each member's strengths and interests, which allowed different subsystems to progress in parallel without significant bottlenecks. Working together in the lab made communication straightforward — issues could be raised and resolved face to face, which meant subsystem integration went relatively smoothly throughout the project. My personal contribution centred on the IR sensing circuit design and the preparation of the interim presentation. For the IR circuit, I designed a signal conditioning stage to detect and distinguish IR pulses reflected from different rock surfaces, iterating through multiple circuit configurations and testing on breadboard. For the presentation, I coordinated input from all subsystems and compiled the slide deck, which required me to engage with each member's work beyond the scope of my own subsystem.

2. Having a team meant that we could support and encourage each other when facing difficulties rather than tackling problems alone. When someone got stuck, others would step in to discuss and brainstorm together, which often helped us look at problems from different angles and arrive at solutions that none of us might have reached individually. For me personally, this was especially valuable when working on the IR circuit. There were moments where I felt stuck and unsure how to proceed, and being able to talk through the problem with teammates helped me think more clearly and consider approaches I had not thought of on my own. Over time, this kind of exchange made me more confident in approaching unfamiliar problems and more open to reconsidering my initial assumptions when things did not go as expected.

3. This project gave me technical and professional skills that I had not fully developed through lectures alone. On the technical side, I was able to apply concepts such as transimpedance amplification, AC coupling, and comparator-based signal conditioning in a practical context, which deepened my understanding of them considerably. I also used LTspice for circuit simulation, which helped bridge the gap between theoretical knowledge and real circuit behaviour.

First Name: Colin
Last Name: Qian
CID: 02556839
Group Number: 14

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1. Our group functioned effectively because each member had a clear responsibility and was able to focus on a specific subsystem. This structure helped us avoid confusion and made progress easier to monitor. As team leader, I contributed by setting development guidelines, deadlines and priorities, while checking whether proposed plans were realistic for our available time, resources and technical ability. I also supported teammates during lab sessions, particularly when debugging or integrating different parts of the robot. My main contribution was keeping the project coordinated and ensuring that individual work combined into one complete system. Over time, the core team developed strong synergy because the team became more cohesive and trusted each other's roles.

2. The diversity of the team improved our performance because members had different technical strengths, working styles and approaches to problem-solving. This allowed us to divide tasks more effectively, such as control, PCB design, mechanical design and testing, according to each person's confidence and ability. It also meant that several ideas could be developed, discussed and compared before we committed to a solution. Different perspectives helped us identify weaknesses earlier, including possible issues in the IR circuit and the integration between hardware and software. When one member was stuck, others could approach the problem differently and help debug it faster.

3. I acquired both professional and technical skills during this project. Professionally, I learned that leadership requires continuous progress monitoring, clear communication, realistic timelines and the ability to stay calm under pressure. I also learned that planning should include backup solutions, since delays or failed designs are common in engineering projects. Managing the team taught me to assign tasks early, match responsibilities to each member's strengths and still give everyone meaningful involvement so they remained motivated. Technically, I learned to use FreeCAD and SolidWorks for 3D-printing design, KiCad for schematic and PCB development, Adafruit coding resources and datasheets for component selection. I also gained experience following a structured engineering process: planning, designing, implementing, testing and evaluating.